



Case study

Enabling better decision-making and optimizing the military workforce

Using automation and artificial intelligence lets the military focus on mission-critical tasks.



1. Better question



The better the question

When data elements are unknown, how can the military find direction?

Using automation and artificial intelligence lets the workforce focus on mission-critical tasks.



Military bases are much more than barracks. Many of them have the populations and infrastructures of small cities, with properties ranging from buildings, fences, water lines, electric lines, parking lots, sidewalks and fire hydrants to cemeteries and hospitals. One military command, which manages the installation's day-to-day operations, needs to track hundreds of thousands of facilities and nearly \$400 billion in assets through its property management systems.

Accurately recording the property is a massive undertaking with vital importance for military decision-making. Simply keeping track of thousands of military buildings, facilities and structures alone would be a difficult task, but the complexity is multiplied by the rigorous accounting that the U.S. Department of Defense (DoD) requires for each individual asset. The DoD requires commands to report and manage over 50 data elements for every one of their assets, making organizations responsible for reporting on millions of data points and there's little room for error. With DoD leadership using property management systems data to create budgets, plan exercises and assess construction needs, accurate data is crucial to the management of the military's real estate.

But these inventories aren't reliable. Inspections have relied on a labor-heavy, inconsistent and decentralized process where inspection information was entered manually into a variety of forms, including Word documents, paper slips, spreadsheets and handwritten notes, that would be saved as pdf documents and stored offline with inconsistent formatting – creating a near-constant risk of user error. Simple checks showed that the military's inventory records didn't accurately reflect reality. Basic information about a building, including its physical condition, square footage or the type of structure, could be incorrect. In one instance, a property listed as a garrison

headquarters actually was a day care center, while other inspections showed that buildings listed as active were boarded up and without running water, or they simply did not exist.

The better the answer

One version of the truth

Through digital tools, the EY team improved the accuracy, reduced the burden on the workforces and provided new tools to military leadership.



In 2019, EY professionals teamed with a DoD property command to streamline processes, aiming to reduce the administrative burden on the workforce and provide military leadership with the accuracy needed for informed decision-making. EY professionals cleaned up the organization's data records and sped up its physical inventory process, saving over 150,000 workforce hours in the first nine months of implementation. The recovered workforce time represented millions of saved taxpayer dollars. By replacing the inconsistent and time-consuming manual process, EY personnel helped optimize the military's workforce and enabled better decision-making among its leadership.

The EY team created consistency in all stages of the inventory documentation process by using an automated management solution that was configured by EY technology consultants. Through a combination of robotic process automation (RPA), optical character recognition (OCR) technology, Power BI and Pega Business Process Management (BPM) software, the EY team created an easily configurable, user-friendly portable solution that delivered a seamless experience so users could see data, access the application and edit information, upload photos or scan documentation to asset records via mobile devices. Through built-in validation, the Pega application forced standardization into data input and eliminated the risk of confusion. Users had to complete all required fields before they could move on. If they tried to enter a four-digit code into a field requiring five digits, they couldn't access the next page, making it clear that they were not adding the required information.

With the property record data collection process cleaned up, RPA technology and OCR software made it easier for the DoD's leadership to put that data into action. Using RPA, a bot took the approved data from the BPM software to update the property records in an enterprise resource planning tool – the central storehouse of property records – enabling an automated process designed to quickly generate reports that accurately reflect data in the field. When the DoD's leadership needed supporting documentation for decision-making, bots and OCR auto-populated the required government forms, replacing an error-prone manual process.

The better the world works

Sprinting past boundaries

Using client feedback and iterative updates, the EY team created a technology solution making asset collection faster and more accurate.



Working with the configurable low-code Pega system, the EY team used the software in a visionary way and pushed the boundaries of its capabilities. Initially, the goal wasn't having it as a complete suite of solutions ready at the start, but instead to offer a viable product that could be tested, expanded and improved as needed during a series of two-week testing and deployment sprints.

The goal was to create a tool any user could pick up and start using immediately, without great risk of error, e.g., using a simple screwdriver, not a table saw. To reach the desired ease of use, the tool was shaped by the experiences of the people using it. The EY team led the designers through iterative updates in a series of two-week sprints, drawing on the feedback from users. Pega's BPM is low code and designed to be easily configured, allowing developers to quickly adapt it to client needs without resorting to hours spent writing complicated code. If users reported that something in the interface was inconvenient, counterintuitive or buggy, that critique would drive

the team's focus for the next two-week sprint update. With each iteration, the tool grew more intuitive.

This story shows how agile thinking and innovations in military tech can unlock unexpected value. Through simplified inventory recording, leadership could redirect its workforce from time-consuming low-level tasks that support the mission's resources to more valuable, mission-critical tasks without needing to hire more people or buy new tools. Previously, personnel spent an average of five hours correcting faulty inventory records. The automated system cut that correction time in half.

Instead of puzzling over unreliable data and inconsistent forms, leadership now can pull up whatever property data is needed, in real time, through the RPI system's dashboard and see what is working and where there are problems. With real-time information about the asset cleanup efforts via Power BI dashboards, military leadership has actionable information about which installations are performing well and which are falling behind.

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